

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

Перенаправление ввода-вывода

```
mrfilosophova@mrfilosophova:~$ ls /etc/ > file.txt
mrfilosophova@mrfilosophova:~$ ls >> file.txt
mrfilosophova@mrfilosophova:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
chrony.conf
cifs-utils
cockpit
colord
containers
```

Рисунок 1: Запись в файл

```
mrphilosophova@mrphilosophova:~$ grep .conf file.txt > conf.txt
mrphilosophova@mrphilosophova:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
locale.conf
logrotate.conf
```

Рисунок 2: Поиск расширения .conf


```
ge02/report/bib/cite.bib  
/home/mrphilosophova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/sta  
ge02/report/bib/cite.bib  
/home/mrphilosophova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/sta  
ge03/report/_resources/csl  
/home/mrphilosophova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/sta  
ge03/report/bib/cite.bib  
/home/mrphilosophova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/sta  
ge04/report/_resources/csl  
/home/mrphilosophova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/sta  
ge04/report/bib/cite.bib  
/home/mrphilosophova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/sta  
ge05/report/_resources/csl  
/home/mrphilosophova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/sta  
ge05/report/bib/cite.bib  
/home/mrphilosophova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/sta  
ge06/report/_resources/csl  
/home/mrphilosophova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/sta  
ge06/report/bib/cite.bib  
/home/mrphilosophova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/  
_resources/csl  
/home/mrphilosophova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/  
bib/cite.bib  
/home/mrphilosophova/git-extended/.git/hooks/commit-msg.sample  
/home/mrphilosophova/git-extended/.git/config  
/home/mrphilosophova/conf.txt  
mrphilosophova@mrphilosophova:~$
```

Рисунок 3: Поиск файлов

```
/etc/foomatic/hashe.s.d/hashe.s.new
find: '/etc/libvirt': Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
find: /etc/nvme/hostnqn
/etc/nvme/hostid
'/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
```

Рисунок 4: Поиск файлов

```
mrfilosophova@mrfilosophova:~$  
mrfilosophova@mrfilosophova:~$ find /etc -name "h*" -print | less  
mrfilosophova@mrfilosophova:~$  
mrfilosophova@mrfilosophova:~$ find ~ -name "log*" > logfile &  
[1] 19157  
mrfilosophova@mrfilosophova:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
mrfilosophova@mrfilosophova:~$ rm logfile  
mrfilosophova@mrfilosophova:~$
```

Рисунок 5: Фоновый запуск процесса

```
mrfilosophova@mrfilosophova:~$  
mrfilosophova@mrfilosophova:~$ gedit &  
[1] 19207  
mrfilosophova@mrfilosophova:~$ ps | grep gedit  
19207 pts/0    00:00:00 gedit  
mrfilosophova@mrfilosophova:~$ kill 19207  
[1]+  Terminated                  gedit  
mrfilosophova@mrfilosophova:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DE(1)                                User Commands                                DE(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount of space available on the file system containing each file name argument.  If no file name is given, the space available on all currently mounted file systems is shown.  Space is shown in 1K blocks by default, unless the environment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a mounted file system, df shows the space available on that file system rather than on the file system containing the device node.  This version of df cannot show the space available on unmounted file systems, because on most kinds of systems doing so requires non-portable intimate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all file systems by default.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        include pseudo, duplicate, inaccessible file systems

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes; see SIZE format below

    --direct
        show statistics for a file instead of mount point

    -h, --human-readable
        Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
du(1)                                User Commands                                du(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=f

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is usually
        smaller, it may be larger due to holes in ('sparse') files, internal fragmentation, indi-
        rect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576
        bytes; see SIZE format below

    -b, --bytes
        equivalent to '--apparent-size --block-size=1'

    -c, --total
        produce a grand total

Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
mrfilosophova@mrfilosophova:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 38859524 109234716 27% /
devtmpfs         4011276      0    4011276  0% /dev
tmpfs            4035648      92    4035556  1% /dev/shm
tmpfs            1614260     1952    1612308  1% /run
tmpfs             1024        0       1024  0% /run/credentials/systemd-journald.service
tmpfs            4035652    132932    3902720  4% /tmp
/dev/nvme0n1p3 155186176 38859524 109234716 27% /home
/dev/nvme0n1p2 1992552    518800    1352512 28% /boot
/dev/loop0        68480     68480        0 100% /var/lib/snapd/snap/core24/1349
/dev/loop1        49280     49280        0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop2       115712    115712        0 100% /var/lib/snapd/snap/hugo/25676
tmpfs             1024        0       1024  0% /run/credentials/systemd-resolved.service
tmpfs            807128     176     806952  1% /run/user/1090
mrfilosophova@mrfilosophova:~$
```

Рисунок 9: Справка по команде du

```
4      ./git-extended/.git/objects/72
4      ./git-extended/.git/objects/9a
4      ./git-extended/.git/objects/55
4      ./git-extended/.git/objects/83
4      ./git-extended/.git/objects/9b
4      ./git-extended/.git/objects/54
76     ./git-extended/.git/objects
8      ./git-extended/.git/logs/refs/heads
8      ./git-extended/.git/logs/refs/remotes/origin
8      ./git-extended/.git/logs/refs/remotes
16     ./git-extended/.git/logs/refs
20     ./git-extended/.git/logs
212    ./git-extended/.git
220    ./git-extended
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
1034052 .
mrfilosophova@mrfilosophova:~$
```

Рисунок 10: Запуск команды du


```
/home/mrphilosophova/git-extended/.git/objects/de
/home/mrphilosophova/git-extended/.git/objects/d6
/home/mrphilosophova/git-extended/.git/objects/72
/home/mrphilosophova/git-extended/.git/objects/9a
/home/mrphilosophova/git-extended/.git/objects/55
/home/mrphilosophova/git-extended/.git/objects/83
/home/mrphilosophova/git-extended/.git/objects/9b
/home/mrphilosophova/git-extended/.git/objects/54
/home/mrphilosophova/git-extended/.git/logs
/home/mrphilosophova/git-extended/.git/logs/refs
/home/mrphilosophova/git-extended/.git/logs/refs/heads
/home/mrphilosophova/git-extended/.git/logs/refs/remotes
/home/mrphilosophova/git-extended/.git/logs/refs/remotes/origin
/home/mrphilosophova/monthly
/home/mrphilosophova/reports
/home/mrphilosophova/reports/monthly
/home/mrphilosophova/reports/monthly/monthly
/home/mrphilosophova/ski.plases
/home/mrphilosophova/ski.plases/equipment
/home/mrphilosophova/ski.plases/plans
/home/mrphilosophova/australia
/home/mrphilosophova/play
/home/mrphilosophova/play/games
/home/mrphilosophova/play/games/play
mrphilosophova@mrphilosophova:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.